

“You said da da...”: A Short Echoing Tweak for Journaling with VA

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ABSTRACT

AI and deep learning have enabled development of voice agents (VAs) and conversational agents (CAs) for more varied uses, including audio journaling which requires longer, more nuanced conversations. Current technology is capable of utilizing additional data, but VAs still remain limited to single-turn, task-oriented interactions. In this study, we investigate whether current turn-taking system is suitable for interactions with long speech in the form of audio journaling, and suggest ways that would make VAs more appropriate to handle longer sentences and turns by the user. To this end, we designed three types of responses, Empathetic, Reading-back, and Mixed, which all echo the user’s speech in different ways. From a week-long journaling experiment with 23 subjects, we collected long speech data (average number of syllables: Empathetic=425, Reading-back=438, Mixed=304). The most effective response in this case is a Reading-back in terms of quality of data and journaling experience.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Voice Agent (VA), turn-taking, multi-turn, interruption, audio journaling

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1 INTRODUCTION

Efforts have been made to enable voice agents (VAs) to communicate with humans in a natural way. One area of research is turn-taking, which detects when the user has finished speaking and responds at the appropriate time. Current VAs typically answer task-oriented responses 1.5 seconds after the user’s speech ends[15]. While this simple turn-taking design is suitable for short, simple single-turn commands, it has limitations when it comes to longer speech. Long speech often includes pauses (mostly longer than 1.5 seconds), which current VAs interpret as the “end” of speech instead of a

“pause”. As a result, the following response may not match the user’s subsequent speech after the pause, resulting in disruptive interruptions.

Journaling is an example of long speech in which frequent pauses occur in the train of thought. It entails recording personal thoughts, daily experiences, and insights. This process can involve conversations with oneself, another person, or even an imagined being[8]. In journaling, the existence of an audience in the monologue is an important factor, and nowadays, the technological development of conversational agents (CA) is creating a new form of journaling. Studies have shown that the presence of a CA can be effective for self-reflection and can enhance engagement with records. Nevertheless, there is limited research on the use of VAs in journaling. According to Helgert’s study on a gratitude diary using a VA, voice inputs and outputs allow for a more intuitive way of writing a gratitude journal. At the same time, participants were under pressure due to VA’s timeout and did not have enough time to write a useful diary entry[7].

In this study, empathy and repetition are applied to the turn-taking of VAs. Empathy-based dialogue systems are particularly effective at reducing negative user emotions, such as aggression[3]. This could help reduce user annoyance caused by interruptions. Additionally, repeating what a user says can prompt them to review their speech and correct any errors[9, 21]. This could help improve the uncertain recording experience due to interruption. In summary, we investigate whether the journaling experience with VAs can be improved through turn-taking responses with empathy and repetition, and if so, whether this can lead to higher quality recorded data.

We developed a VA that helps record a diary about the previous day in reverse chronological order. To see the difference in turn-taking responses, we designed three versions: Empathetic, Reading-back, and Mixed. The Empathetic agent reflects emotional empathy for the user’s speech. The Reading-back agent repeats the last two words of the user’s speech. Lastly, the Mixed agent reflects these two conditions. Through this, we examined whether there is an effect of the agent’s turn-taking response on the long speech journaling experience where pauses often occur.

A 7-day in-situ user study was conducted with 23 subjects randomly assigned to three conditions. Each subject wrote a diary about the previous day in reverse chronological order. The findings show that VAs were able to record long speeches: 425 syllables per record on average in the Empathetic turn-taking condition, 438 syllables in the Reading-back condition, and 304 syllables in the Mixed condition. Among them, the most effective turn-taking design in terms of data and experience was the Reading-back condition. The data collected under the Reading-back condition consistently maintained data quantity and quality, unlike the other conditions. Additionally, subjects showed the highest annoyance

level over inappropriate empathetic responses, and reading-back responses significantly improved the transparency of recording experience. This is a preliminary study using VAs in the corresponding use case, and proposes appropriate turn-taking design principles in a long speech situation by providing empirical understanding in an in-situ environment.

2 RELATED WORK

In general, people have negative perceptions about interruption. However, in order to collect conversation data using the current VA, interruption is inevitable, and therefore it is necessary to turn the experience into a positive one. In this study, we would like to find out the design principles of positive turn-taking in the case of audio journaling.

2.1 Improved Turn Taking with Interruptions

Currently, VAs judge that the speech is over after 1.5 seconds of end-of-speech time and attempt turn-taking for security[15, 17]. However, this study suggests that end-of-speech may not always indicate the "end of speech" and should be interpreted in more diverse contexts. If it is not the end of the speech, i.e. if it is turn-taking in multi-turn rather than single-turn, handling it only once is not enough. The experience itself, which will happen multiple times, should be improved to avoid disrupting users who try to record long enough speech.

Sacks referred to the case where the listener violates the appropriate timing of turn-taking and speaks overlapping with the current speaker's speech to take control of the conversation as a negative "interruption"[18, 22], and the current VA's turn-taking in response to a long speech acts as an interruption due to end-of-speech timeout. However, Bennett argued that some interruptions may exist as 'intervention' in a positive or neutral sense[1]. "Intervention" in this case serves to increase the efficiency of communication by intentionally helping or responding to the words, even though the listener overlapped with the current speaker's speech. Therefore, if the VA's turn-taking is improved from interruption to intervention, it becomes possible to receive a long speech from the user.

2.2 Improved Audio Journaling with Interruptions

Audio journaling is much simpler than traditional hand-written journaling and can capture real-time experiences, providing rich life-logging data[14]. This results in an unstructured narrative, with frequent unintended pauses. According to the audio diary records collected in Lynn's research, there are frequent pauses (.) between unfinished sentences. He also found that audio journaling often takes the form of a conversation, even without a physical audience[13]. When there is an audience, speakers can feel a strong sense of companionship and engage with the records, as if they are sharing experiences with the audience in real time through conversations with a researcher[2, 16].

The dialogue principles of positive intervention, such as empathy and repetition, are used to foster an emotional bond between conversation participants, rather than efficient communication. This gives users more agency, rather than the system-driven agency[4, 10]. These responses demonstrate that the system is listening to the

user's speech, and encourages more conversations by promoting intimacy. It also provides a connection with what has been said, allowing users to continuously check and correct their speech and gradually proceed, resulting in a continuous multi-turn, rather than the existing single-turn of a mere command/execution[5, 19].

Therefore, the research questions in this study are presented as follows:

RQ1: Can turn-taking with empathy and repetition receive long speech data in journaling with VAs?

RQ2: Can turn-taking with empathy and repetition improve the user experience in journaling with VAs?

3 METHOD

3.1 System Design

In this study, we developed a VA that can be used in Google Assistant using Google Dialogflow and Google Spreadsheet. It has three different turn-taking responses: Empathetic, Reading-back, and Mixed. When a turn-taking occurs, it tries to empathize with the result of sentiment analysis of the user's speech (e.g. "You might be happy/sad/surprised/angry...") or provide a hint about the speech recognition state of the VA through reading-back (e.g. "I heard up to da da..."). See Figure 1 for more details.

For empathy, an emotion analysis model was trained using AI HUB's 'conversational dataset containing Korean emotion information'¹. The data consists of 28594 sentences labeled with 7 emotions: joy, sadness, surprise, anger, fear, disgust, and neutrality. For repetition, the last two words of the previous recorded speech are repeated.

3.2 Participants and Task

Initially, 30 people responded to the study's recruitment notice, of which 23 (18 female, 5 male; mean age=27.30, SD=18.13) successfully finished the study. Seven participants were excluded during the study due to technical errors such as microphone malfunctions or severe sore throats caused by COVID-19. To avoid learning effects, the study was conducted with a between-subjects design. Group 1 used an Empathetic agent, Group 2 used a Reading-back agent, and Group 3 used a Mixed agent.

Thirteen subjects were heavy users of VAs, using them more than once a week, while ten subjects were light users, using them more than once a month. To ensure familiarity with the VAs, heavy and light users were equally distributed across all groups.

For natural empirical understanding, the study was conducted for a week in the subjects' daily spaces. Each day, they verbally recorded a diary in reverse chronological order about the previous day, irrespective of time and space. As the reverse chronological diary requires the reconstruction of time along with recall, more pauses occur in the audio journaling process. During recording, the subjects were instructed to turn their displays upside down to remove any visual cues. After recording, they completed a survey and edited the recorded text. The survey was conducted with the same VA used for recording, and text editing was done with Google

¹https://aihub.co.kr/reti_data_board/language_intelligence

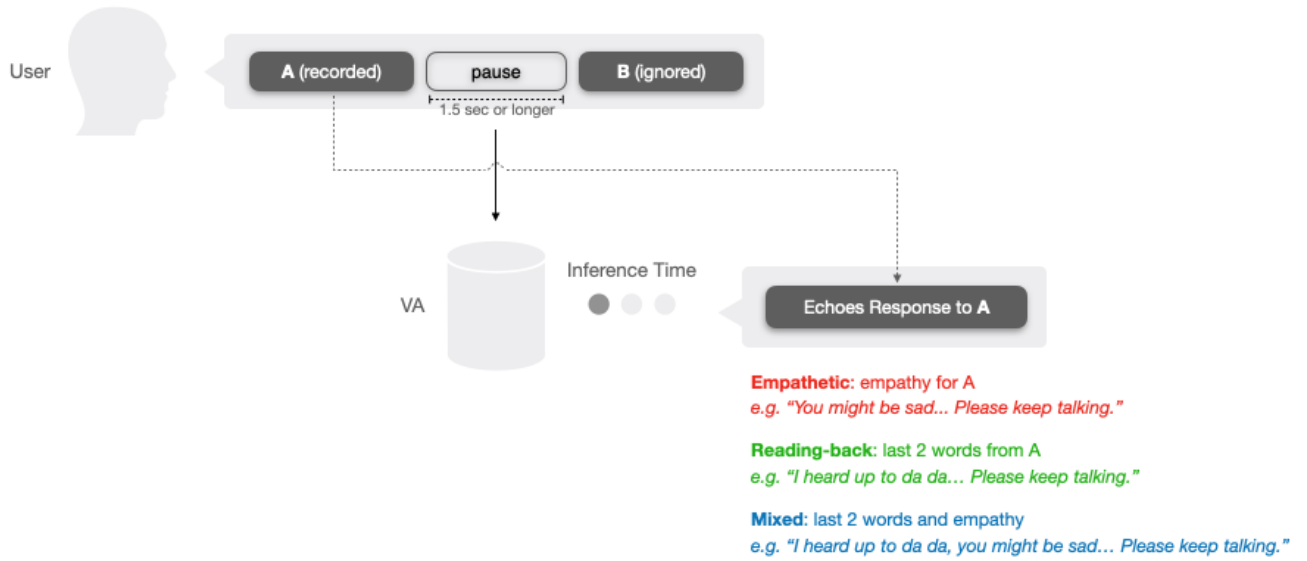


Figure 1: When VA detects a pause of 1.5 seconds or longer, it recognizes the speech before the pause (A), takes some time to infer intent, handle data, etc., and then responds. Without any visual cues, users cannot know the system of turn-taking, so they may perceive it as an interruption and feel annoyed. Additionally, the speech after the pause (B) is not recorded by the VA, but users are unaware of this. Therefore, as an appropriate turn-taking response, we designed Empathetic, Reading-back, and Mixed responses.

Appsheet, which is linked to Google Spreadsheet. After the week-long in-situ study, post-interviews were conducted to gain a deeper understanding of their experiences.

3.3 Procedure

The study procedure consisted of system and task guides, task performances, and interviews. Before the task period, researchers provided guides for the system and the task. During the task period, participants then completed a set of {record-survey-edit} tasks once a day. After the task period, a 30-minute elicitation interview was conducted with the researchers via Zoom. Upon completion of all these procedures, participants received \$40 as an honorarium.

3.4 Measure and Analysis

The **journaling data** consisted of the initial data recorded through VAs and the final data modified through a web app.

- The **attain rate** was calculated by multiplying the rescaled qualitative score (ranging from 0 to 1) by the quantitative length of the initial journaling data. The qualitative score was assigned by two researchers on a 5-point scale, considering the completeness ("Is every event from night until morning described in detail?") and plausibility ("Do the preceding and following sentences flow logically?") of the content. The researchers then finalized their agreement to achieve an inter-coder reliability of at least 80%[12]. ANOVA was used to analyze the attain rate in order to confirm RQ1.
- The **edit rate** was calculated by dividing the amount of edited text (excluding typos) by the length of each final

journaling data. During the interview, we checked whether each editing context was due to typos from voice recognition or not. ANOVA was used to analyze the edit rate in order to confirm RQ2.

Surveys captured the evaluation of annoyance[20] over interruption after each recording, with answers collected on a 5-point scale to "how annoyed were you by the interruption in accomplishing what you were asked to do?" ANOVA was used to analyze the scores in order to confirm RQ2.

Interviews were conducted at the end of the study. Participants were asked about the reasons behind their survey scores and the context for any edits they made, which supported the statistical results from the quantitative analysis of the data. In addition, they were asked about their overall experience and feedback on audio journaling with VAs. The answers were transcribed and analyzed using grounded theory techniques[6].

4 RESULTS

As a result of the study, a total of 160 sets of records, survey scores, and edited data were collected. One set of data was lost due to technical errors. Since differences in records were very large for each subject, we analyzed statistical data and interviews together with mixed-methods.

4.1 Quantitative Findings

4.1.1 VA's turn-taking can receive a long speech. The average attain rate for subjects (a in Figure 2) in the Empathetic condition was 382.92 (SD=173.53, n=49), in the Reading-back condition it was

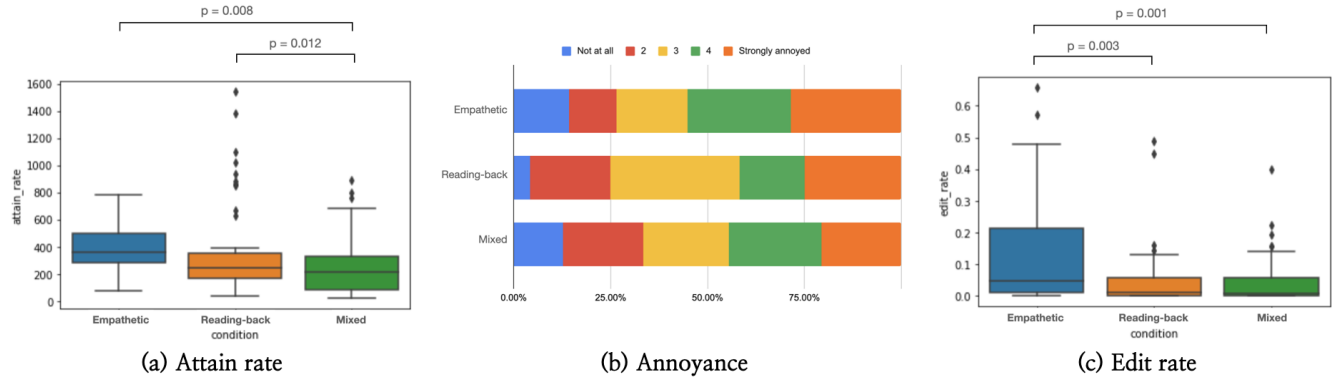


Figure 2: Attain rates, 5-point Likert-scale ratings on annoyance, and edit rates under each condition are compared by (a), (b), and (c) respectively.

377.73 (SD=357.53, $n=48$), and in the Mixed condition it was 237.67 (SD=192.33, $n=63$). A one-way ANOVA and post-hoc Tukey test revealed significant differences between the Empathetic condition and the Mixed condition (HSD=-145.26, $p=0.008$), as well as between the Reading-back condition and the Mixed condition (HSD=-140.06, $p=0.012$). This suggests that subjects created a longer and more devoted journaling data under Empathetic and Reading-back turn-taking, while creating a shorter and less devoted record under the Mixed one. The related qualitative results are discussed in Section 4.2.1.

4.1.2 Reading-back turn-taking can improve the experience of recording. Regarding annoyance (b in Figure 2), subjects scored an average of 3.43 points (SD=1.39, $n=49$) in the Empathetic condition, 3.38 points (SD=1.18, $n=48$) in the Reading-back condition, and 3.19 points (SD=1.32, $n=63$) in the Mixed condition. One-way ANOVA revealed no significant difference among the three conditions, indicating that the type of turn-taking responses had no effect on improving annoyance. We will discuss the related qualitative results in Section 4.2.2.

Regarding the edit rate (c in Figure 2), subjects edited records at an average ratio of 0.13 (SD=0.17, $n=49$) in the Empathetic condition, 0.05 (SD=0.10, $n=48$) in the Reading-back condition, and 0.04 (SD=0.07, $n=63$) in the Mixed condition. One-way ANOVA and post-hoc Tukey test showed a significant difference between the Empathetic condition and the Reading-back condition (HSD=0.08, $p=0.003$), as well as between the Empathetic condition and the Mixed condition (HSD=0.09, $p=0.001$). This suggests that subjects recorded accurately and made fewer edits through turn-taking that repeats the last two words. We will discuss the related qualitative results in Section 4.2.3.

4.2 Qualitative Findings

4.2.1 Reading-back turn-taking provides an appropriate strategy for recording using VA. During the study, almost every subject ($n=21$) learned a recording strategy for turn-taking while using the system for a week. They either attempted to 1) prevent the VA from interrupting by speaking continuously without pauses, or 2) allow the

VA to intervene by speaking relatively briefly with some pauses. In the Empathetic condition, most subjects tended to adopt the former strategy, whereas subjects in other conditions were more likely to learn the latter one. This was also consistent with the average speech length, which refers to the average number of syllables per sentence (a in Figure 3).

The reason why the Empathetic condition led to the first strategy was that the subjects had difficulty noticing the system of turn-taking: “Initially, my words were being interrupted, and sometimes not even recorded. To avoid further errors, I started speaking without stopping.” (P22 - Group 1) One subject even ignored the response of turn-taking due to its wrong empathy response: “I wasn’t particularly conscious of the response, I just quickly said what I had to say and finished recording.” (P19 - Group 1)

In the Reading-back and Mixed conditions, participants were assisted in figuring out the timing of turn-taking through a reading-back response. This encouraged them to regulate pauses during speech. However, adding an empathy response resulted in complaints such as: “I felt annoyed and distracted when the agent strangely empathized with me while I was trying to think of what to say next.” (P15 - Group 3) and “The agent’s response was too long and made me uncomfortable as I had to wait silently until it was over. It felt like the main and sub were switched.” (P9 - Group 3)

Using turn-taking effectively during speaking had a positive impact on engagement in conversation with a VA. Participants who learned to use turn-taking with intentional pauses enjoyed and were satisfied with the journaling experience. However, those who did not know how to use turn-taking found it disruptive and tried to reduce the number of turns taken by speaking non-stop, as shown by the average number of turns taken (b in Figure 3). P11 described the moment when he and the VA’s turn-taking got along well: “I think the timing of the end of my speech and that of his intervention matched well, so I was able to continue the conversation satisfactorily.” (P11 - Group 2)

4.2.2 There are adverse effects of incomplete empathy. Reasons why there was no significant difference in annoyance under each condition in 4.1.2 are as follows.

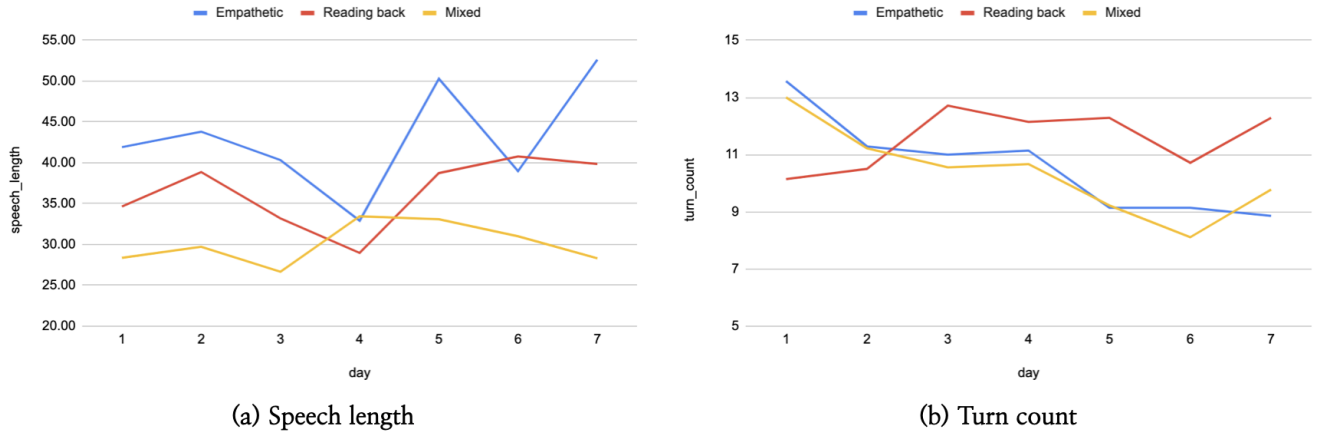


Figure 3: Changes in speech length and turn count under each condition are shown by (a) and (b) respectively.

Firstly, inappropriate empathetic responses added to annoyance. 9 out of 16 subjects who heard the empathetic turn-taking response raised questions about the empathy that did not fit the context and said that they could not find any difference from the machine-like repetition response. There was even a negative perception about a neutral empathetic response that was not in the wrong context: *"I thought it might be a bit of forced reaction because he reacted like 'that's right' (neutral empathy) every time."* (P18 - Group 3)

Secondly, external factors were taken into account. For example, the first conversation partner of the day showed incomplete empathy, which aggravated the user's annoyance: *"I was more annoyed than the day before because I had a conversation with the agent for the first time in the morning without talking to anyone."* (P3 - Group 3)

Thirdly, as time went on, the annoyance levels were downgraded under all conditions since the users adapted to a wrong turn-taking response of the VA. Adaptation was found in two contexts, either being disappointed at the machine or feeling closer to its social traits. P10 was disappointed and said, *"As I got used to it, I felt less annoyed. Maybe it was my fault to expect him to react like a human being, he's just a machine anyway..."* (P10 - Group 1). P12 felt affection and said, *"I could feel the agent's effort to respond. It was cute. It was like a less educated kid."* (P12 - group 3).

4.2.3 Reading-back turn-taking organizes speech. Based on the elicitation interview, participants made two types of edits to recorded text, excluding typos resulting from speech recognition errors. First, they made edits because the recording status was unclear. For instance, they added words that were not recorded, even though they had spoken them before, or deleted duplicate words because they spoke multiple times without knowing whether it had been recorded. Second, they made edits because their thoughts were directly verbalized in words. Only after visually checking the results of their spoken words could they add, delete, or move words to ensure contextual completeness and plausibility.

It is clear that using reading-back turn-taking responses, which confirm the recording status by repeating the last two words of the speech, can reduce the need for the first type of edits. P9, who

had not experienced reading-back turn-taking response, said, *"I re-recorded short sentences from the beginning, and long sentences from two or three words before the interruption."* (P9 - Group 1). On the other hand, P8, who had experienced reading-back turn-taking response, said, *"It was good to write a verbal diary because I can notice how far I have spoken."* (P8 - Group 2).

However, during the interview, it was discovered that the second type of edits occurred less frequently in the Reading-back and Mixed conditions. This was mostly due to participants being able to structure their thoughts in advance by intentionally pausing. In contrast, participants in the Empathetic condition spoke without pauses and later organized their thoughts, resulting in more edits: *"I felt that what I said in words suddenly jumped in terms of content."* (P9 - Group 1)

5 DISCUSSION AND CONCLUSION

This study explored the feasibility of recording long speech through VA. The turn-taking system of VA, which was most problematic for this research question, was improved with conversational principles of empathy and repetition. A week-long journaling study was conducted under three conditions to test the improved VA. The results showed that it was possible to record long speech through the improved VA, and we investigated this in more detail in terms of data and experience. Through reading-back turn-taking response, it was possible to consistently collect a rich quantity and quality of journaling data. While empathetic turn-taking response could be rather annoying, reading-back turn-taking response provided a transparent recording experience.

5.1 Turn-taking Design Implications in Multi-turn

The results indicate that machine-like repetition can enhance the comprehension of opaque turn-taking systems. As users gain a better understanding of the turn-taking system, they are more likely to utilize pauses effectively. This finding contradicts prior research, which had focused on receiving single-turn speech from users and processing it intelligently with a sense of autonomy. Instead, this

study suggests that providing users with an understanding of the system via transparent responses (similar to Explainable AI[11]) is necessary to improve the experience of long speeches where turn-taking occurs multiple times. In multi-turn and journaling situations, the user should have agency and continuous engagement, rather than the system taking control.

5.2 Advantages of Audio Journaling with VA

Partly due to turn-taking, there has not been much research on audio journaling using VAs. However, we could discover a number of advantages by conducting a preliminary empirical study. First, the modality makes journaling quicker and easier. Subjects who had been diligently writing a diary praised the fact that they did not have to use their hands to write a large amount of text. Second, self-reflection occurs more effectively than with handwriting a diary. They said that a verbal conversation with turn-taking that happens periodically made them feel like they were talking to themselves. Finally, it can serve as an emotional outlet. Subjects said that the agent enabled them to freely express even minor emotions, since they are usually conscious of or burdened by others.

5.3 Limitation and Future Work

One potential limitation of this study is the lack of a control turn-taking response (e.g. "Yes, and?"), which makes it difficult to compare the results to a baseline. This study suggests that recording long speeches through VA's turn-taking is possible, as all participants tended to speak more in those multi-turns compared to using general VA in a single-turn. However, it is unclear whether these results were solely due to the improved turn-taking responses or if other factors played a role. Further research is needed to validate this. Another limitation is that the efficacy of empathetic turn-taking response in reducing annoyance has not been adequately verified. As detailed in Section 4.2.2, subjects still exhibited some degree of social traits even when conversing with agents that did not fully empathize. Hence, if social presence is further enhanced through an agent capable of expressing rich empathy similar to humans, a different human-like turn-taking strategy may emerge. Moreover, there are many use cases for long speech with frequent pauses other than journaling, such as dream records, medical history taking, interview machines, etc. VAs can be used in various scenarios, and as their intelligence advances, it will be possible to handle long speeches in more diverse situations through further research.

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